

Approaching person

Moving in the world frame: **the person**. Staying in place: the railing, the tree, the sign, 2 doors.

The person moves about 1.1 m in the world frame at an average speed of 0.68 m/s.

Question

Right answer: A

In the world frame (ignoring camera motion), what was the average speed (in meters per second) of **the person** between frame 0 and frame 6 of 26?

A. 0.52 B. 0.95 C. 0.70 D. 1.3

ground truth: 0.59 m/s

Qwen3-VL-8B · frames only

```
<evidence>
object: the person
position_frame_0_m: (0.8, 1.6, 6.5)
position_frame_6_m: (0.4, 1.2, 4.8)
displacement_m: 1.8
time_s: 1.20
speed_mps: 1.53
</evidence>
<answer>D</answer>
```

Ours · frames + trajectory tokens

```
<evidence>
object: the person
position_frame_0_m: (0.8, 0.5, 4.8)
position_frame_6_m: (0.3, 0.6, 4.3)
displacement_m: 0.7
time_s: 1.20
speed_mps: 0.58
</evidence>
<answer>A</answer>
```